

Inverse Design of Diffractive Metasurfaces Using Diffusion Models

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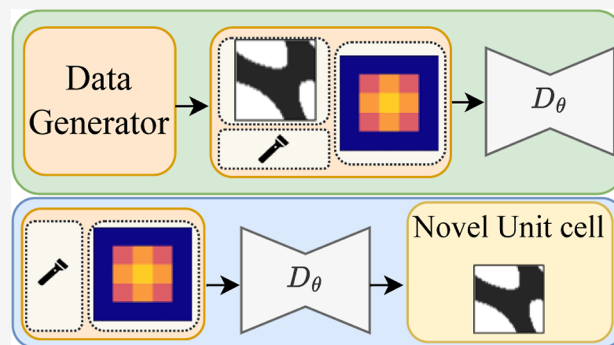
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ABSTRACT: Metasurfaces are ultrathin optical elements composed of engineered subwavelength structures that enable precise control of light. Their inverse design—determining a geometry that yields a desired optical response—is challenging due to the complex, nonlinear relationship between structure and optical properties. This often requires expert tuning, is prone to local minima, and involves significant computational overhead. In this work, we address these challenges by integrating the generative capabilities of diffusion models into computational design workflows. Using an RCWA simulator, we generate training data consisting of metasurface geometries and their corresponding far-field scattering patterns. We then train a conditional diffusion model to predict meta-atom geometry and height from a target spatial power distribution at a specified wavelength, sampled from a continuous supported band. Once trained, the model can generate metasurfaces with low error, either directly using RCWA-guided posterior sampling or by serving as an initializer for traditional optimization methods. We demonstrate our approach on the design of a spatially uniform intensity splitter and a polarization beam splitter, both produced with low error in under 30 min. To support further research in data-driven metasurface design, we publicly release our code and data sets.

KEYWORDS: metasurfaces, diffusion models, inverse design, nanophotonics



1. INTRODUCTION

Diffractive metasurfaces are optical devices composed of subwavelength meta-atoms that enable precise control over light propagation. Designing these structures to achieve specific optical properties remains a significant challenge due to the complex and nonlinear relationship between the geometry and the optical response.

Early research on metasurface inverse design focused on classical optimization techniques. In particular, gradient-based methods relying on numerical light-scattering simulations, such as rigorous coupled-wave analysis (RCWA),¹ were extensively explored.^{2–4} However, these approaches often converge to local minima. To address this, alternative strategies, such as topology optimization, evolutionary algorithms, and particle swarm optimization, have been proposed.^{5,6} These are frequently combined with global phase initialization schemes, such as the iterative Fourier transform algorithm (IFTA), to improve convergence.^{7–9} While effective, these methods are computationally intensive, often requiring hours to design a single metasurface, and depend on expert-tuned regularization strategies.

With the rise of deep learning, neural networks (NNs) have also been explored for metasurface design.^{3,10–13} However, these models often underperform analytical methods due to their reliance on large, task-specific data sets which are rarely available in practice.

It is important to note that most prior work on metasurface inverse design using NNs has focused on the spectral response as the primary design objective. In contrast, controlling the far-field spatial power distribution, namely the directionality and intensity of scattered Bragg orders, has received limited attention, despite its importance in beam splitting, shaping, and multiplexing applications. Recent diffusion-based studies have also explored metasurface inverse design.^{11–13} However, refs 11 and 13 target spectral responses (S-parameters), while our work addresses far-field *spatial* power control across discrete Bragg orders. Although ref 12 also considers spatial responses, it is restricted to parametrized unit cells (i.e., not freeform) and focused on the RF spectral band. Second, we emphasize scalability and reproducibility of the data used for learning, aiming for broad structural variety rather than fixed curated sets. Furthermore, prior works typically enforce the desired response only during training, whereas we also enforce it at generation time (details provided below), which we find improves robustness beyond the training distribution. The key distinctions are summarized in Table 1.

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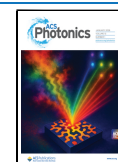


Table 1. Concise Comparison with Prior Diffusion-Based Inverse Design Works

aspect	this work	prior diffusion-based works ^{11–13}
design objective	far-field <i>spatial</i> power distribution over Bragg orders	mainly targets <i>spectral</i> response; spatial in RF with restricted parametrization
data	free-form, generated per task	fixed data sets with narrower variety
conditioning	during training and generation	during training only

We next outline our problem setting and high-level approach, which is schematically presented in Figure 1. This paper addresses the inverse design of metasurfaces that produce a desired spatial far-field power distribution. We consider periodic, binary-thickness dielectric metasurfaces under normal incidence. Since the wavelength and geometric features are of comparable scale, the periodicity induces scattering into discrete diffraction orders. Our goal is to recover a meta-atom configuration that yields a prescribed angular power distribution at specified wavelengths.

To this end, we have developed a generative framework based on diffusion models^{14,15}—a class of neural generative algorithms increasingly used to generate high-fidelity samples and solve inverse problems. We train the model on a large data set of simulated structure–response pairs, constructed using a novel, scalable data generation pipeline. It should be emphasized that in contrast to the more common tasks tackled by NNs, such as image classification, generation or text prediction, a large training data set does not exist and must be constructed ad hoc. Our data generation method addresses the lack of high-quality task-specific data and can be adapted to various nanophotonic design problems.

Once trained, diffusion models can be sampled to generate solutions that match the learned distribution. We explore two operational modes:

- **RCWA guidance (posterior sampling):** full-wave simulations (e.g., RCWA) are integrated into the diffusion generative process, trading off accuracy vs runtime.

- **Integrative mode:** the diffusion model provides initial guesses for downstream optimization routines.

It should be noted that the training process (especially the data set generation method—see Section 2) creates a distribution of topology and diffraction pattern pairs. It can be expected that some of the metasurface topologies (and, hence, the attainable diffraction patterns) are not well-represented within this distribution. Consequently, the design of metasurfaces targeting such diffraction patterns is challenging for our diffusion model. We tackle this gap through incorporation of careful considerations to the data generation process and state-of-the-art methods for querying diffusion models for desired measurements consistency.

We demonstrate the effectiveness of our approach on several challenging design tasks, particularly targeting patterns which are not well-represented within the learned distribution. These include the generation of SiO₂ metasurfaces exhibiting an X-shaped scattering pattern across multiple wavelengths, the design of a silicon dual-polarization uniform beam splitter, and a polarization splitter refined via gradient descent, all within minutes.

This paper is organized as follows: Section 2 details the data generation process, the inverse design process, and evaluation methodology. Section 3 presents experimental results and demonstrates the usage modes of our diffusion model for metasurface inverse design.

2. METHOD

In this section, we present our proposed method for the inverse design of diffractive metasurfaces using diffusion models. We begin by introducing a novel data generation procedure. We then describe the training of our diffusion model, which we designate as *MetaGen*, and how it is used to design new metasurfaces. Finally, we outline our evaluation methodology, which includes comparisons with alternative generative frameworks and the design of metasurfaces corresponding to target scattering patterns. Additional implementation details are provided in Sections 2–4 of the Supporting Information.

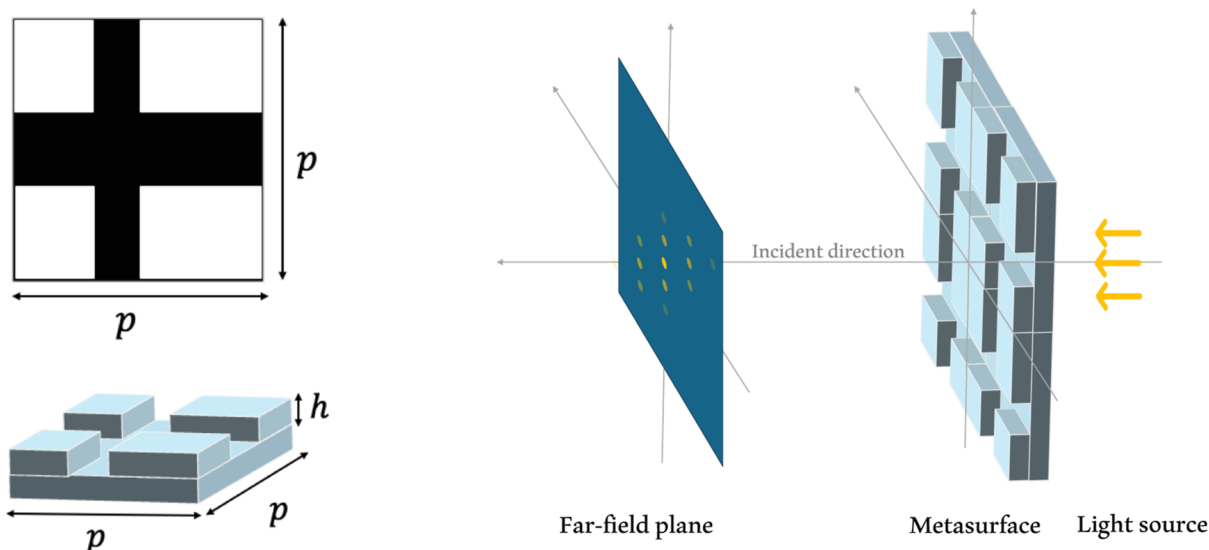


Figure 1. A schematic illustration of the metasurface setup: (left) a meta-atom as a binary image and its 3D schematic, and (right) the overall scattering scenario with plane-wave excitation.



Figure 2. Binary structures from our proposed data set. Different gray levels represent different thicknesses h . By design, the structures are naturally periodic and continuous.

2.1. Data Generation. The data sets consist of pairs of binary images that represent the topology of the metasurface and the transmitted and reflected scattering efficiencies T and R , which are represented as matrices. Each pixel in these matrices contains the total power projected into one of the Bragg diffraction orders, where the central pixel corresponds to the zero-order diffraction lobe.

2.1.1. Binary Structures Randomized Generation. To construct a diverse data set of binary metasurface structures, we developed a custom generation pipeline that produces meta-atoms from randomized noise. The process begins by applying a low-pass spatial filter to realizations of uniformly sampled noise $\sim \mathcal{U}(0, 1)$. While the random sampling promotes diversity and free-form variation, the low-pass filter enforces a minimum feature size, ensuring compatibility with standard fabrication constraints. We perform the spatial filtering in the frequency domain using a 2D Fourier transform, which naturally enforces periodic boundary conditions in the generated meta-atoms. The filtered noise is then binarized, where each resulting binary pattern encodes a metasurface design: 1 denotes dielectric material, and 0 denotes air. To further diversify the data set, we generate additional parametrized shapes, such as rectangles, ellipsoids and grating profiles—structures which may not naturally arise from free-form generation. For implementation details, see Section 2 of the Supporting Information. Selected free-form structures are presented in Figure 2.

2.1.2. Simulations. The light scattering patterns from the metasurfaces are computed using rigorous coupled-wave analysis (RCWA),¹ a numerical method for solving Maxwell's equations in periodic structures by representing electromagnetic fields as truncated Fourier series. We use a GPU-accelerated, differentiable implementation of RCWA.² In each data set, the metasurface thickness and operating wavelength vary uniformly across samples, while other physical parameters, such as the periodicity and material composition, are fixed. All simulations are carried out under normal incidence, though this assumption can be relaxed by generating data sets corresponding to oblique illumination conditions.

2.1.3. Datasets. To demonstrate our method under varying physical conditions, we constructed two main data sets, B2 and C2. Both data sets are designed to exhibit at least 5×5 propagating orders in air, but differ in the material, the supported wavelengths and the periodicity. The number of propagating orders was controlled by selecting the periodicity relative to the operating wavelengths, considering that each material is transparent only within its characteristic transmission band. For more details, please refer to Section 2.1 in the Supporting Information. Additional data sets with a different number of excited orders are detailed in Section 2 in the Supporting Information. Both data sets consist of 3.6 M (720 K data points per operating wavelength), with the resolution of the binary images set to 64×64 pixels.

2.1.4. Consideration of Fabrication Constraints. To ensure that our designs remain compatible with contemporary fabrication capabilities, we explicitly account for minimal feature size limitations during data generation. Specifically, as mentioned above, we apply spatial low-pass filtering in the randomization stage of structure creation, guaranteeing that features below the allowed size are suppressed. For data set B2, features smaller than ~ 290 nm are removed, while for C2 the cutoff is ~ 460 nm. Since binarization can occasionally reintroduce subresolution features, we further apply an iterative smoothing–binarization procedure, which both enforces the minimal feature size and induces smoother structural curvatures. This approach ensures that the generated data sets correspond to designs that are not only physically meaningful but also manufacturable with current nano-fabrication techniques.

2.2. Inverse Model. We train a diffusion model, *MetaGen*, to serve as an inverse predictor, reconstructing the geometrical attributes of a metasurface—specifically, its binary structure and height—given a target spatial power distribution and an operating wavelength λ . The model outputs a 2D binary image representing the meta-atom layout, along with a corresponding scalar height value.

Training is performed separately on the B2 and C2 data sets (see Table 2) using a standard diffusion modeling approach.

Table 2. Physical Settings of the Main Datasets^a

data set	B2	C2
polarizations	TE only	TE & TM
material	SiO ₂ ($n \approx 1.45$)	Si ($n \approx 3.6$)
wavelengths λ [μm]	{0.8, 0.85, 0.9, 0.95, 1.0}	{1.4, 1.45, 1.5, 1.55, 1.6}
heights h [μm]	0.75–1.25	0.1–0.6
periodicity p [μm]	2.86	4.6

^aThe B2 and C2 datasets were generated using our method and are used for training. Both are designed to ensure the presence of scattering patterns with at least 5×5 propagating Bragg orders, by choosing the periodicity relative to the operating wavelengths. Additional data sets and configurations are described in the Supporting Information.

Samples from the training set are corrupted with additive Gaussian noise, and the model is trained to denoise them while conditioned on the corresponding target scattering patterns and wavelengths. Through this process, the model learns the statistical relationship between metasurface geometry and its optical response. The training procedure follows the elucidated diffusion models (EDM) framework,¹⁶ with further details provided in Sections 1 and 4 of the Supporting Information.

2.2.1. Sampling. Once trained, *MetaGen* can be sampled through an iterative process to generate valid metasurface designs starting from pure noise. See Figure 3 for a schematic overview. To further enforce consistency with physical

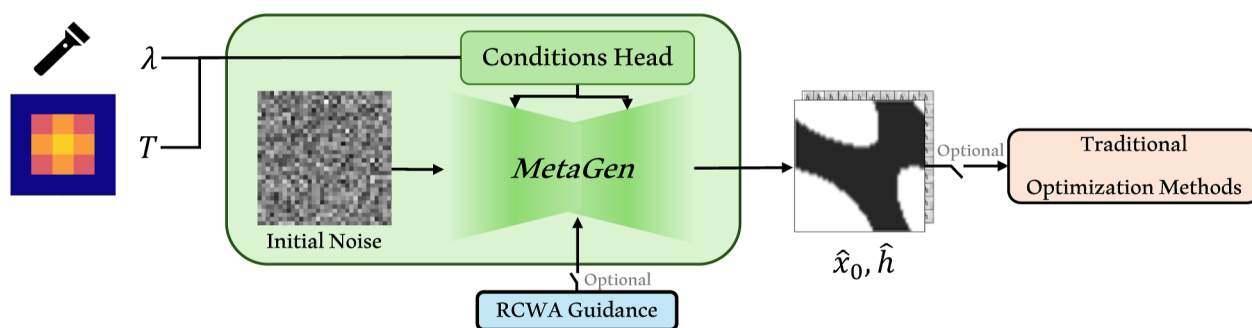


Figure 3. *MetaGen*: diffusion-based inverse design. We train a diffusion model to reconstruct the geometry of a metasurface, specifically, the binary pattern x and height h , starting from pure noise and conditioned on the target far-field scattering pattern and operating wavelength. The model's output is denoted by $\hat{x}_0, \hat{h}$, following the conventional notation used in diffusion models.

measurements, we employ diffusion posterior sampling (DPS).¹⁷ This technique integrates gradients from a forward simulator (RCWA in our case) into the diffusion trajectory, guiding the generation toward samples that are both physically plausible and consistent with the target scattering pattern. In our context, we refer to this approach as RCWA guidance. This technique requires backpropagating through the simulator, which increases computational cost in exchange for improved accuracy. In practice, we apply RCWA guidance either at all diffusion steps or only at selected ones, depending on the complexity of the target design task. We note that the model can also be sampled directly without RCWA guidance, enabling near-instant generation. While this unguided approach performs well on target patterns drawn from the training distribution, we observe reduced effectiveness when generalizing to idealized or out-of-distribution target diffraction patterns.

2.3. Evaluation. We evaluate model performance by comparing the target scattering patterns T, R , which condition the generation process along with the wavelength λ , to the actual scattering patterns $\hat{T}, \hat{R}$ obtained from simulating the predicted meta-atom structures using the RCWA simulator. Given its relevance in many applications, our evaluation primarily focuses on the relative error in transmittance (T). When assessing performance across multiple components (e.g., two polarizations), the relative error is computed for each component separately and then averaged.

2.3.1. Metrics. We propose using relative error as the metric, as it intuitively represents the relative amount of power that is either missing or incorrectly distributed between the diffraction lobes compared to the desired scattering pattern. The error metric is defined as follows for $\mathbf{x} \in \{T, R\}$

$$e(\mathbf{x}, \hat{\mathbf{x}}) = \frac{\|\mathbf{x} - \hat{\mathbf{x}}\|_1}{\|\mathbf{x}\|_1} \quad (1)$$

2.3.2. Evaluation Strategy. Since *MetaGen* relies on the data distribution used during training, it is important to evaluate its performance not only on the test set, drawn from the same distribution as the train set, but also on manually constructed scattering patterns representative of real-world applications. We generally refer to such scattering patterns as target patterns. In Section 3, we present three such application-driven designs obtained using our approach. To further assess the effectiveness of diffusion models in capturing the training distribution, and also generalizing to target patterns beyond the observed distribution, we compare *MetaGen* with other generative baselines. Specifically, we evaluate a modified

Wasserstein generative adversarial network (GAN) with gradient penalty (WGAN-GP),^{18–21} adapted for conditional sampling, and a conditional variational autoencoder (CVAE)^{22,23} designed for similar inverse tasks. A detailed comparison is provided in Section 3 of the Supporting Information.

2.3.3. Visualization. Scattering patterns are shown as heat maps, with each Bragg order (m_x, m_y) color-coded by intensity. Values above 0.01 are reported with two decimal places.

3. RESULTS

This section presents results demonstrating the effectiveness of *MetaGen* in solving real-world optical design tasks. First, we present its broad spectral support by generating metasurfaces that produce scattering patterns that are beyond the learned distribution across multiple wavelengths. More specifically, we show the ability of *MetaGen* to design a metasurface exhibiting X-shaped scattering patterns. Next, we showcase the design of a spatially uniform scattering pattern, achieving performance comparable to prior works but with significantly reduced runtime. Finally, we demonstrate the use of *MetaGen* to generate candidate metasurface designs that serve as initializations for auxiliary optimization procedures, thus reducing significantly the design time. As a concrete example we demonstrate fast design of a metasurface based polarization beam splitter.

As illustrated in Figure 3, the sampling process begins from a random noise realization, which gradually transforms into a fresh sample valid under the training distribution, i.e., a metasurface. Thus, the generation process is essentially stochastic, and multiple candidate metasurfaces may be designed by initializing the sampling process with different noises. We leverage this variability, together with the parallelism capabilities of modern GPU-accelerated frameworks, to improve design outcomes with only marginal runtime overhead. For each design task presented in this study, we generate k candidate samples in parallel and select the one that yields the lowest scattering error.

3.1. Wide Spectral Support and Wavelength Variation Robustness. We use *MetaGen* to generate a nontrivial (out of the learned distribution) X-shaped scattering pattern across a range of wavelengths: $\{0.825, 0.875, 0.925, 0.975\}$ $[\mu\text{m}]$. Note that none of these wavelengths is present in the B2 training set. This demonstrates the model's broad spectral support and its ability to interpolate between trained wavelengths. This design is conducted under the B2 configuration, i.e., SiO_2 based metasurface with TE-polarized

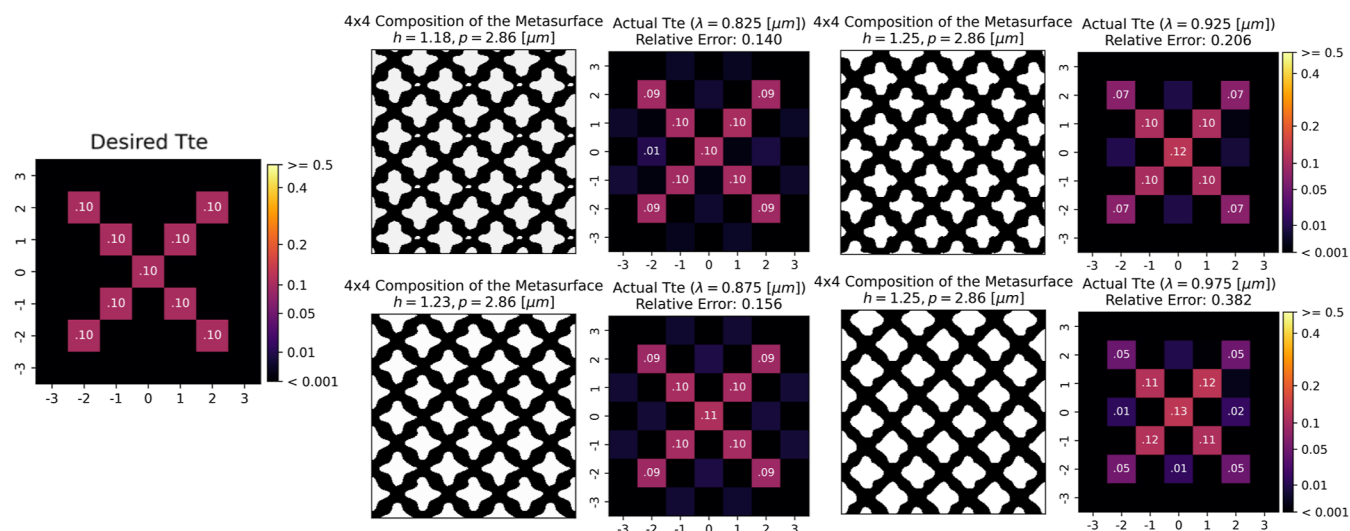


Figure 4. Wide spectral support. Our model is trained on multiple discrete wavelengths, enabling it to generalize to unseen wavelengths through interpolation. In this example, we design metasurfaces that produce an X-shaped scattering pattern across several operating wavelengths outside the training distribution.

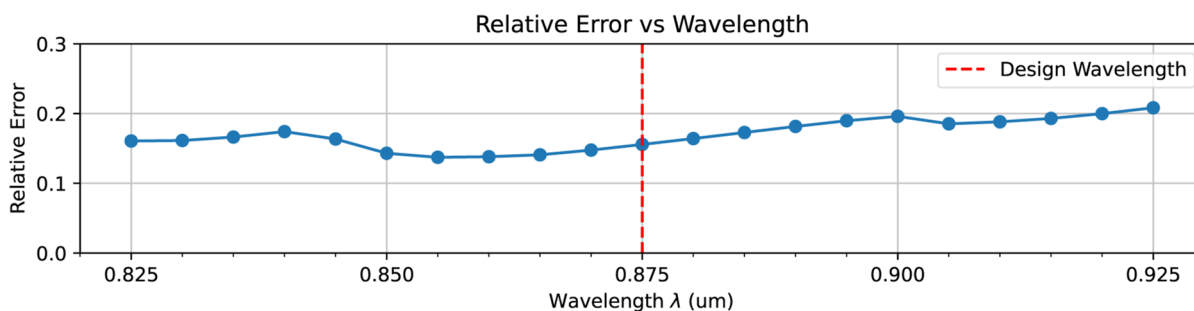


Figure 5. Robustness to wavelength variations. Relative error of the metasurface designed for $\lambda = 0.875 \mu\text{m}$ when evaluated at different operating wavelengths. The results show that the device performance remains stable under moderate wavelength shifts.

incident light only (see Table 2). Figure 4 shows the resulting metasurface designs (for each wavelength) and the actual scattering patterns they produce.

As a complementary analysis, we evaluated the robustness of the designed devices to variations in the operating wavelength. Specifically, we selected the metasurface designed for $\lambda = 0.875 \mu\text{m}$ (Figure 4, bottom-left design), calculated its diffraction pattern across a range of operating wavelengths, and evaluated the relative error. The results, shown in Figure 5, demonstrate that even substantial wavelength shifts lead to only minor deviations in the device response.

3.2. Spatially Uniform Intensity Beam Splitter. In this section, we demonstrate the design of a beam splitter, i.e., a metasurface that uniformly distributes the intensity of the incident plane wave across a 5×5 Bragg diffraction orders. Prior studies^{4,7–9} have addressed this task using optimization procedures that typically require several hours. Our goal is to significantly accelerate the design process while maintaining comparable performance. In contrast with prior works, we choose to showcase a design that is simultaneously constrained to exhibit a uniform scattering pattern in both polarizations. This represents an extension of the original challenge, which typically considers only a single polarization mode. Following these works, we evaluate our design using the uniformity error (UE) metric

$$e_{\text{UE}}(\hat{T}) = \frac{T_{\text{max}} - T_{\text{min}}}{T_{\text{max}} + T_{\text{min}}}, \quad \text{where} \quad T_{\text{max}} = \max_i T_i, \\ T_{\text{min}} = \min_i T_i \quad (2)$$

where $\{T_i\}$ denotes the transmitted power in each diffraction order.

The results are summarized in Table 3. Our model, trained under the C2 configuration, achieves comparable performance

Table 3. Inverse Design of a 5×5 Uniform Beam Splitter^a

method	Hao et al. ⁷	Li et al. ⁸	Yan et al. ⁹	MetaGen (ours)
UE (eq 2)	0.1740	0.1264	0.1176	0.1230/0.1600
runtime	3.5 h	2.6 h	18.9 h	28 min

^aCompared to prior works, our method achieves comparable UE while completing in a fraction of the runtime. The runtime for Hao et al. was estimated from the reported duration of 90 s per generation and visual inspection of figures (approximately 140 generations). Runtime comparisons may also be affected by hardware differences; see Section 6 of the supporting information for full specifications.

to those of previous works while simultaneously constraining both polarizations. The corresponding design is shown in Figure 6. Notably, our solution was generated in just 28 min using a best-of- $k = 30$ sampling strategy, in contrast to the hours-long runtimes reported in previous works. To achieve

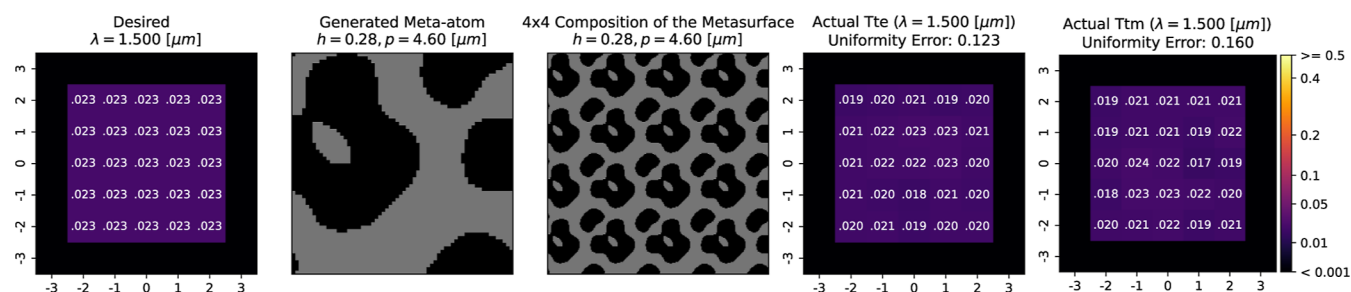


Figure 6. Uniform beam-splitting metasurface. Our approach integrates information from RCWA measurements directly into the diffusion sampling process. This allows us to overcome local-minima sensitivity without expert tuning and to achieve high-quality results within minutes. In this example, we present a uniform, dual-polarization 5×5 beam-splitting silicon metasurface designed for normally incident light at wavelength $\lambda = 1.5 \mu\text{m}$.

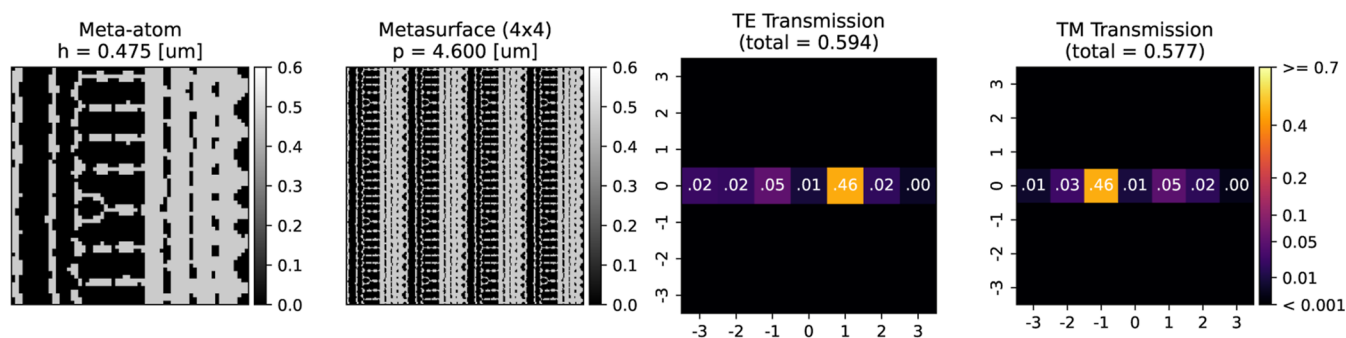


Figure 7. Polarization splitter design using *MetaGen*. Candidate patterns generated by *MetaGen* were used as initializations for gradient descent optimization, consistently outperforming randomly initialized baselines. Here, 800 optimization steps were performed with a decaying learning rate, and the objective function was the mean squared error (MSE). Each pixel contains the fractional amount of transmitted power.

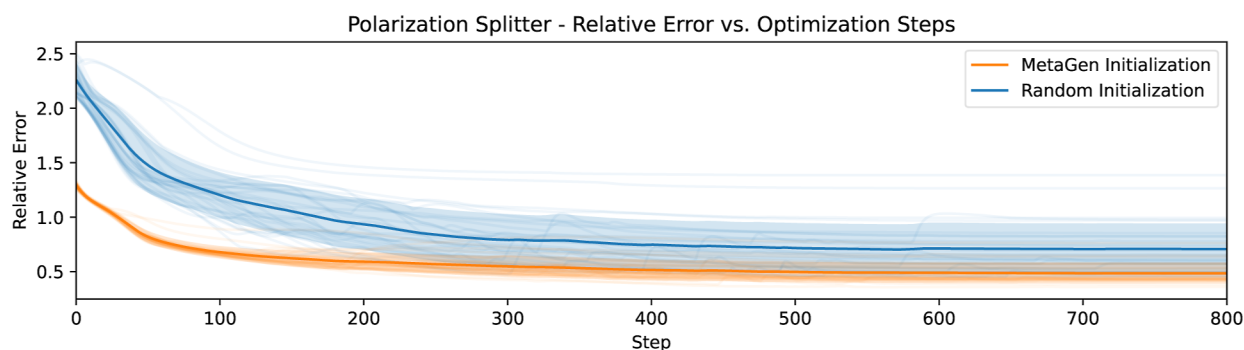


Figure 8. Optimization performance for polarization splitting. The blue and orange curves show the average relative error across 30 randomly and model-based initialized runs, respectively. The shaded regions denote the standard deviation.

this, we use 200 diffusion steps and apply RCWA-based guidance during the final 100 steps.

3.3. Polarization Splitter: Integrating Traditional Optimization. Recent studies on the inverse design of diffractive metasurfaces commonly employ conventional optimization techniques, initialized randomly or using phase-response methods such as IFTA. Given the vast design space and susceptibility to local minima, population-based algorithms are frequently used to improve convergence.^{7,9} While effective, these approaches are typically computationally intensive and require expert-tuned initialization through IFTA.

We propose leveraging *MetaGen* as an expert-free, data-driven initializer within such optimization pipelines. By providing good initial guesses, *MetaGen* mitigates local-minima convergence issues and significantly accelerates the overall design process. As a reference method, we adopt gradient descent (GD) with smoothing and gradual projection steps, as

outlined in.^{2,4} The smoothing step involves applying a spatial low-pass filter to suppress high-frequency features, while the gradual projection step progressively enforces binarization. Further details on the GD-based implementation for metasurfaces inverse design are provided in Section 5 of the Supporting Information.

In this example, we target the design of a polarization beam splitter, a device that directs TE- and TM-polarized incident waves into different diffraction orders. We begin by querying *MetaGen*, which was trained on dual-polarized Silicon data set C2 (see Table 2), to generate patterns that aim to split both polarizations evenly between the (1, 0) and (−1, 0) diffraction orders. These patterns are then used as an initialization for GD, which is applied to refine the design and enhance the scattering response.

Figure 7 presents an example of a polarization splitter generated using this hybrid approach. Our results show that

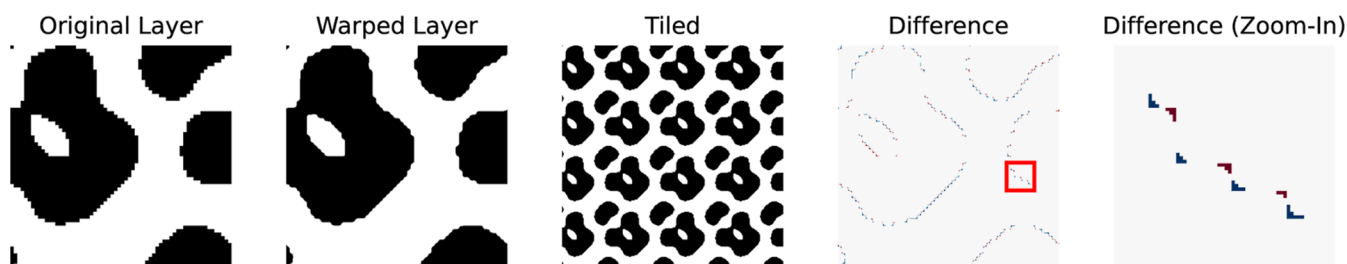


Figure 9. Illustration of fabrication-induced perturbations. The panel on the right highlights the differences between the original and warped layouts, Blue pixels correspond to expansion with respect to the original layer, whereas red pixels indicate shrinkage.

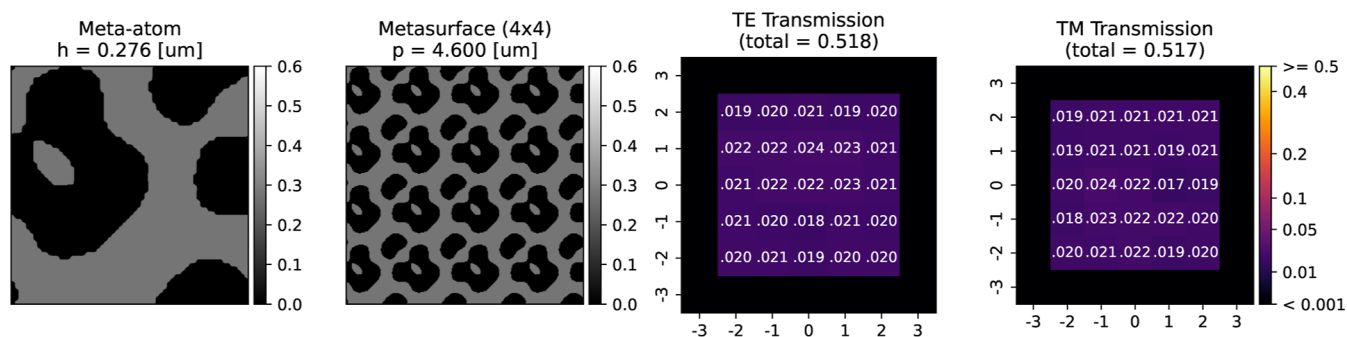


Figure 10. The actual scattering pattern of our uniform beam splitter (Section 3.2) under displacement perturbations, modeling standard fabrication errors.

initializing auxiliary optimization methods such as GD with outputs from *MetaGen* consistently improves final performance, as illustrated in Figure 8. Moreover, the reduced variance of model-initialized optimization trajectories could also benefit population-based methods, such as genetic algorithms and particle swarm optimization.

In the hybrid approach, sampling from *MetaGen* is performed directly, i.e., without applying RCWA guidance, thus completes almost instantly. The subsequent optimization process requires approximately 8 min on the hardware used in this study. Overall, high-quality metasurface designs can be generated within a matter of minutes. Further details regarding the computational setup are provided in Section 6 of the Supporting Information.

3.4. Additional Results and Baseline Comparisons.

For quantitative evaluation, we compare our diffusion model with other generative approaches, which we also train on data sets constructed using our data generation method. We provide WGAN-GP and C-VAE baselines on additional data sets (C1–C3) of increasing structural complexity. As detailed in Supporting Information Section 3, *MetaGen* consistently achieves lower relative error on both test and target patterns (Table S2, Figure S6), demonstrating superior accuracy and generalization beyond the training distribution. Visual results further support its advantage in modeling complex structure–response mappings (Figures S4–S7). Notably, these findings align with prior works, in which diffusion models were consistently proven to outperform GANs also in other applications.^{24,25}

3.5. Robustness to Fabrication Errors. Standard fabrication processes can introduce small deviations from the intended design. While thickness variations are typically negligible, spatial placement errors can slightly warp the geometry of the unit cell. To model such imperfections, we apply controlled warping to our finalized designs and re-evaluate their target objectives.

As a concrete example, we perturb the Uniform Beam Splitter design from Section 3.2 30 times, each by a different random pixel-level shift, either expanding or shrinking local features. The perturbations are done in 512×512 resolution for 1–2 pixels, such that for the data set C2, they translate to 9–18 nm shifts. Under these shifts, the UE (eq 2) of the splitter changes on average from 0.123/0.160 (TE/TM) to 0.128/0.163, exhibiting degradation in one component and improvement in the other, both within an acceptable error margin. Figure 9 shows an example of a perturbation and its effect.

In order to further show the insignificance of these slight displacement error, we present the perturbation that cause the worst degradation of the UE metric of all the 30 randomized experiment. See Figure 10.

Another possible fabrication error is nonuniform thickness. However, current nanofabrication technologies achieve uniformity with nanometer-level precision, making this effect relatively small. For completeness, we analyze the potential impact of such errors on devices designed with our approach in Section 3.4 of the Supporting Information.

4. CONCLUSION

In this work, we introduce a diffusion model-based framework for the inverse design of metasurfaces conditioned on specified spatial scattering patterns across multiple incident wavelengths. Our method can operate in a standalone manner to generate metasurfaces with low error. Additionally, accuracy can be improved at the cost of increased computation time by integrating a RCWA simulator into the sampling process, enabling posterior refinement. Furthermore, recognizing the limitations of traditional optimization methods, which often require expert-tuned initialization or incur long runtimes, we demonstrate how our model can serve as a fast, expert-free initializer to accelerate and improve the performance of such methods. Our diffusion model is trained on paired data

consisting of metasurface structures and their corresponding scattering patterns. To support this approach, we develop a novel, scalable data generation pipeline capable of producing training data at any scale, for a wide range of physical settings and tasks. We construct two primary data sets encompassing diverse physical conditions to enable systematic evaluation of inverse design methods. By designing applicable optical devices, such as a uniform beam splitter and a polarization beam splitter, we show that our diffusion model achieves performance comparable to state-of-the-art approaches that typically require hours of optimization, while significantly reducing computational overhead. These results highlight the potential of diffusion models for scalable and efficient metasurface design, opening avenues for further research in data-driven nanophotonics. To foster reproducibility and support further progress in the field, we release our models, data sets, and code at <https://github.com/liavhen/metagen>.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsp Photonics.5c01384>.

An extended introduction to diffusion models and their role in solving inverse problems, a detailed description of the data generation process, further evaluation results comparing our method to WGAN-GP and CVAE baselines, and implementation details for all models used in this study (PDF)

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Notes

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